

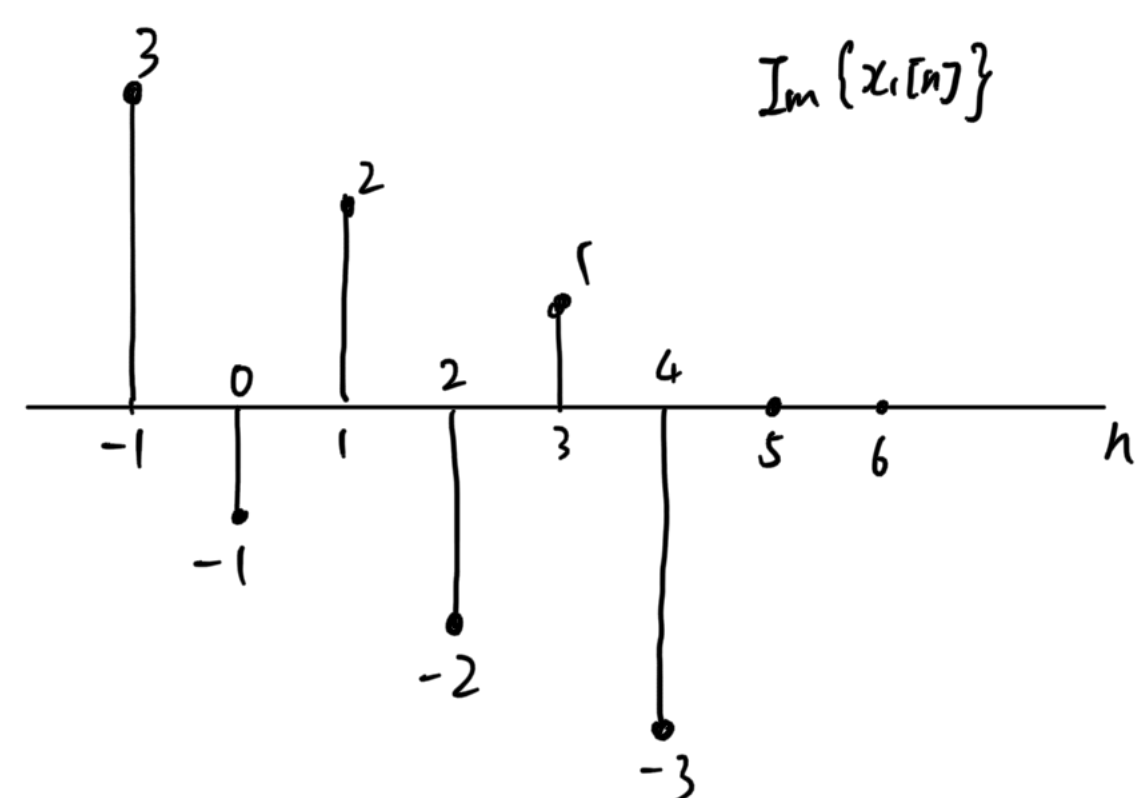
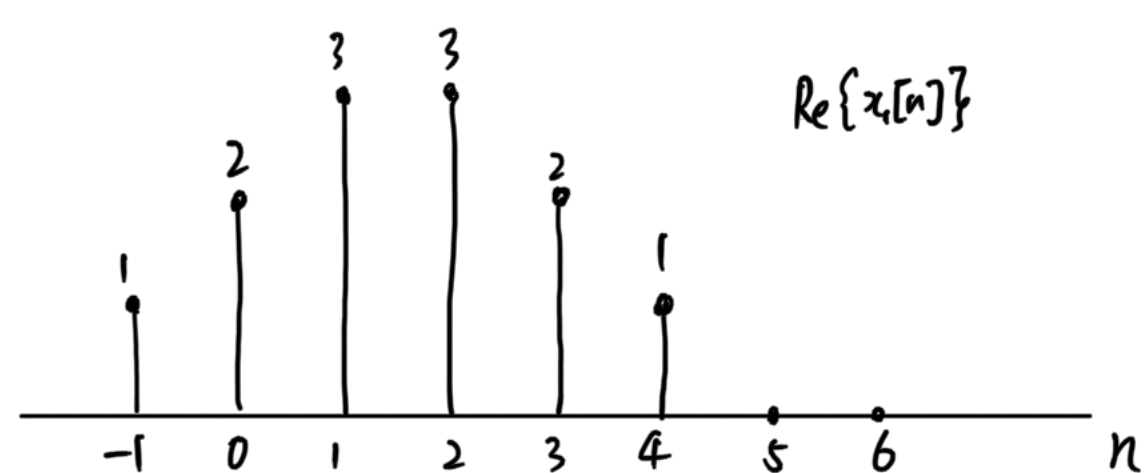
$$2.44 \quad (a) \quad X(e^{j\omega}) \Big|_{\omega=0} = \sum_{n=-\infty}^{+\infty} x[n] = 12$$

$$(b) \quad X(e^{j\omega}) \Big|_{\omega=\pi} = \sum_{n=-\infty}^{+\infty} x[n] (-1)^n = -12i$$

$$(c) \quad \int_{-\pi}^{\pi} X(e^{j\omega}) d\omega = 2\pi x[0] = 4\pi - 2\pi i$$

$$(d) \quad X(e^{-j\omega}) = \sum_n x[n] e^{j\omega n} = \sum_n x[-n] e^{-j\omega n}$$

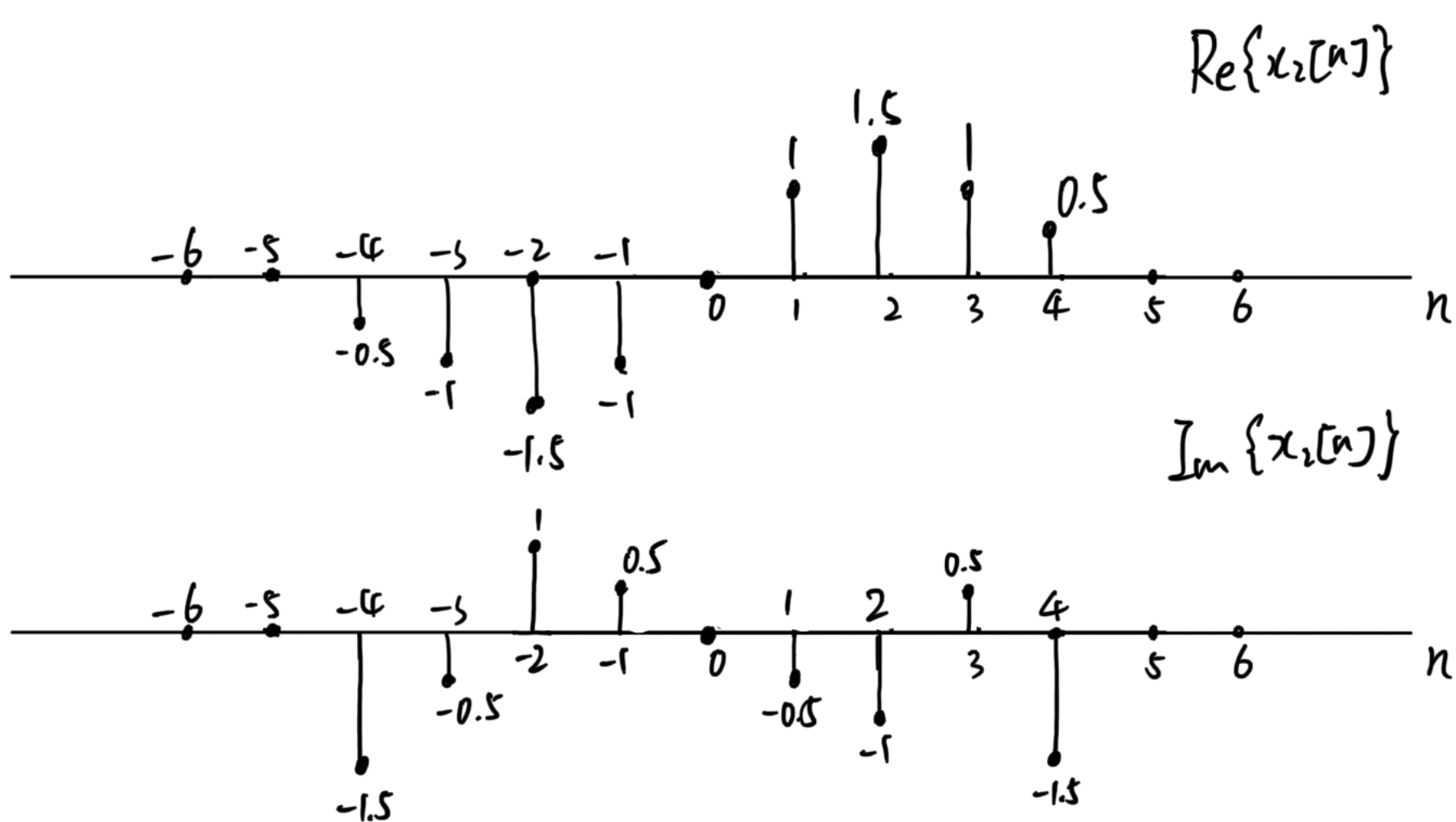
所以 $x[n] \leftrightarrow X(e^{j\omega}) \Leftrightarrow$ 所求信號 $x_1[n] = x[-n] \leftrightarrow X(e^{-j\omega})$



$$(e) \quad x[n] = x_o[n] + x_e[n], \quad X_o(e^{j\omega}) = j \text{Im}\{X(e^{j\omega})\}$$

$$\text{所以 } x_o[-n] \leftrightarrow X_o(e^{-j\omega}) = j \text{Im}\{X(e^{j\omega})\}$$

$$\text{所求信號 } x_1[n] = x_o[-n] = \frac{x[-n] - x[n]}{2} = \frac{x_e[n] - x_e[-n]}{2}$$



$$2.89 \quad (a) \quad \Phi_{xx}[m] = E\left[\sum_{n=-\infty}^{+\infty} x[n] x[n+m]\right] = \Phi_{ss}[0] + \Phi_{ee}[0] + E\left[\sum_{n=-\infty}^{+\infty} s[n] e[n+m]\right] + E\left[\sum_{n=-\infty}^{+\infty} e[n] s[n+m]\right]$$

$$= \Phi_{ss}[0] + \Phi_{ee}[0] + \Phi_{se}[m] + \Phi_{es}[m]$$

$$\begin{aligned} \Phi_{xx}(e^{j\omega}) &= \sum_{m=-\infty}^{+\infty} \Phi_{xx}[m] e^{-j\omega m} = (\Phi_{ss}[0] + \Phi_{ee}[0]) \cdot 2\pi \sum_{k=-\infty}^{+\infty} \delta(\omega - 2k\pi) + \Phi_{se}(e^{j\omega}) + \Phi_{es}(e^{j\omega}) \\ &= (\Phi_{ss}[0] + \Phi_{ee}[0]) \cdot 2\pi \sum_{k=-\infty}^{+\infty} \delta(\omega - 2k\pi) + 2\text{Re}\{\Phi_{es}(e^{j\omega})\} \end{aligned}$$

$$(b) \quad \Phi_{xe}[m] = E\left[\sum_{n=-\infty}^{+\infty} x[n] e[n+m]\right] = \Phi_{se}[m] + \Phi_{ee}[m]$$

$$\Phi_{xe}(e^{j\omega}) = \Phi_{se}(e^{j\omega}) + \Phi_{ee}(e^{j\omega})$$

$$(c) \quad \Phi_{xe}[m] = E\left[\sum_{n=-\infty}^{+\infty} x[n] s[n+m]\right] = \Phi_{ss}[m] + \Phi_{es}[m]$$

$$\Phi_{xe}(e^{j\omega}) = \Phi_{ss}(e^{j\omega}) + \Phi_{es}(e^{j\omega})$$

$$2.92 \quad (a) \quad \sigma_x^2 = E[x[n]^2]$$

$$E[x[n]y[n]] = E[x[n] \sum_{m=-\infty}^{+\infty} h[m] x[n-m]] = \sum_{m=-\infty}^{+\infty} h[m] R_{xx}[m] = h[0] \sigma_x^2$$

$$(b) \quad E[y[n]] = E[x[n]] \sum_{k=-\infty}^{+\infty} h[k] = 0$$

$$\begin{aligned} \sigma_y^2 &= E[y[n]^2] = E\left[\left(\sum_{m=-\infty}^{+\infty} h[m] x[n-m]\right)^2\right] = E\left[\sum_{m=-\infty}^{+\infty} \sum_{k=-\infty}^{+\infty} h[m] h[k] x[n-m] x[n-k]\right] \\ &= \sum_{m=-\infty}^{+\infty} \sum_{k=-\infty}^{+\infty} h[m] h[k] R_{xx}[m-k] \\ &= \sum_{m=-\infty}^{+\infty} h[m] \left(h[m] * R_{xx}[m]\right) \\ &= \sigma_x^2 \sum_{m=-\infty}^{+\infty} h[m] h[m] = \sigma_x^2 \sum_{m=-\infty}^{+\infty} h[m]^2 \end{aligned}$$

$$2.96 \quad (a) \quad E[x^2[n]] = \Phi_{xx}[0] = \frac{1}{2\pi} \int_{-\pi}^{\pi} \Phi_{xx}(e^{j\omega}) d\omega$$

$$(b) \quad \Phi_{xx}(e^{j\omega}) = \Phi_{ww}(e^{j\omega}) \Phi_{hh}(e^{j\omega})$$

$$= \sigma_w^2 |H(e^{j\omega})|^2$$

$$= \frac{\sigma_w^2}{1.25 - \cos\omega}$$

$$(c) \quad H(e^{j\omega}) = \frac{1}{1 - 0.5e^{-j\omega}} \Leftrightarrow h[n] = 0.5^n u[n]$$

$$\Phi_{xx}[n] = \Phi_{ww}[n] * h[n] * h[-n]$$

$$= \sigma_w^2 \delta[n] * h[n] * h[-n]$$

$$= \sigma_w^2 h[n] * h[-n]$$

$$= \sigma_w^2 \sum_{k=-\infty}^{+\infty} h[k] h[k-n]$$

$$= \sigma_w^2 \sum_{k=0}^{+\infty} 0.5^k u[k] \cdot 0.5^{k-n} u[k-n]$$

$$= \sigma_w^2 \sum_{k=\max(0,n)}^{+\infty} 0.5^{2k-n}$$

$$= \frac{\sigma_w^2 0.5^{2\max(0,n)-n}}{1 - 0.25}$$

$$= \frac{\sigma_w^2 0.5^{|n|}}{0.75}$$